



Guardrails are safety mechanisms that control what goes into and comes out of an AI agent. They sit around your agent pipeline and ensure the agent:

- Only processes safe, appropriate inputs
- Only performs approved actions
- Only returns validated, compliant outputs

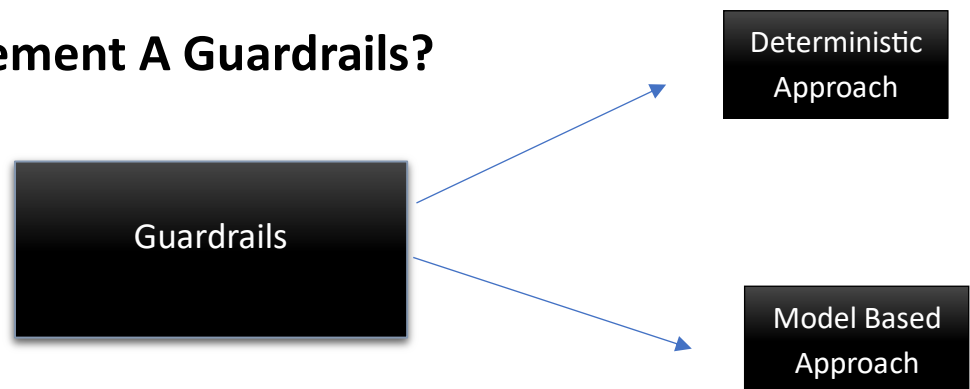
This is a basic AI agent, it's actually doing it taking an input. The input goes to the LLM. Then LLM makes a call either to the tools or it can also directly give the output. Now whenever I talk about these tools, these tools can be RAG database, vector database, it can be APIs, it can be MCP server, anything as such and also packages. But if it's making a call to the tool, here we get a specific context. And then LLM combines this context along with the prompt and then the output is generated.

Now when what about the Guardrails!!

User gives an input "Hey, how to hack a server?" The input is not appropriate. My LLM does that either it flags the content saying that is not good or before the input going to the LLM, some checks should happen here, saying that the input is flagged and this is not appropriate input, so not generated any output. So the output is generated by this AI Engine always is validated and

correct and based on rules and regulation . If I give a image to the input and tell the llm that generated a image ,and make all these things or swap the face from the particular person's face. So that's not look good for this we implement Guardrails.

How to implement A Guardrails?



In Model based Approach :we use here LLMs and also semantic meaning clearly understood by the LLM .It is easy if I give a not validate input and telling the LLM to flag or not flag based on specific prompt so LLM definitely understand the semantics .The here you able to see that easily any kind of violations can be mentioned to the LLM ,so that it catches those kind of violations and stops the request then and there itself.

In this model based approach there also disadvantages is that LLM is costly and cost of every Api call .

In the deterministic Approach: it is Rule based like regex ,keyword matching and also here no llm so no llm cost ,but you don't understand the semantic

LangChain Uses In Implement Guardrails:

By **Lang chain** uses I Implement the guardrails ,lang chain has the important ways Of handling this, in the form of middleware ,middleware-within the agent workflow we can add different kinds of hooks within the workflow ,before the

agent after the agent .consider the middleware they are three six important steps .

1.PII Middleware :

- Inbuild middleware in Lang chain that help to detect email ids ,credit card ,numbers, IPs, URLs. And within the agents, you can integrate this kind of property so that it makes sure that it tells the agent to probably restrict output or try to apply some other kind of validations.
- Whenever we use PII Middleware it uses some kind of techniques like Masking ,It provides an Hash ,it is an algorithm that changes the entire numbers that is given or in the email Ids that is given over there.
- It also applies to the i/p, o/p and also applies to the tools calls and this all about PII Middleware

2. Human In the Loop Middleware :

- It pauses agent before any sensitive tools and It will wait from the human either approve or reject.
- Also here you have to implement threads and checkpoints , so that it understand for which user we are trying to talk .

3. Before Agent Hook:

- It runs before any LLM call
- Say that the guardrails is validated or not
- It is zero cost for blocked requests ,Because we are not hitting the LLM .So what we are doing is that we are blocking them over there is no cost basically involved with the LLM. And then if it getting blocked we can directly move this into the end state. It is so amazing.It probably goes and sees that mechanism before the LLM call and sees that, okay if that getting flagged ,it is just directly going to send to the end workflow or end of the AI agent.

4.After Agent Hook:

- Where we can specifically apply the guardrails is after the agent hook. Let's say the agent has executed, it has generated the output or final response and after that also you will be able to validate before user sees it .So let's say before user sees it, if there is some kind of flag it needs to do, it will be able to do.
- It can also replace or muted unsafe content .
- Also you can use here cheap LLM model in order to implement this.

5. Layerd Guardrails: here you can combine everything whatever mentioned over here ,combine it in the form of stacks and go ahead and implement it